

COS80023 Big Data – Lab 6 – Credit Task

Lab 6: Credit Task 6 – Messaging with Kafka

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Zookeeper Instance Started

```
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\zookeeper-server-start.bat config\zookeeper.properties
[2025-09-24 23:20:33,761] INFO Reading configuration from: config\zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
```

```
apache.zookeeper.server.ServerCnxnFactory)
[2025-09-24 23:20:33,854] INFO binding to port 0.0.0.0/0.0.0.0:2181 (org.apache.zookeeper.server.NIOServerCnxnFactory)
```

Kafka Broker Started

```
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-server-start.bat config\server.properties
[2025-09-25 00:57:29,029] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration$)
[2025-09-25 00:57:29,266] INFO starting (kafka.server.KafkaServer)
```

```
[2025-09-25 00:57:30,115] INFO Kafka version: 2.3.0 (org.apache.kafka.common.utils.AppInfoParser)
[2025-09-25 00:57:30,115] INFO Kafka commitId: fclaaa16b661c8a (org.apache.kafka.common.utils.AppInfoParser)
[2025-09-25 00:57:30,115] INFO Kafka startTimeMs: 1758725850109 (org.apache.kafka.common.utils.AppInfoParser)
[2025-09-25 00:57:30,117] INFO [KafkaServer id=0] started (kafka.server.KafkaServer)
```

Creating topics streams-plaintext-input and streams-wordcount-output.

```
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-topics.bat --create --topic streams-plaintext-input --bootstrap-server 127.0.0.1:9092 --partitions 1 --replication-factor 1
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-topics.bat --create --topic streams-wordcount-output --bootstrap-server 127.0.0.1:9092 --partitions 1 --replication-factor 1
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-topics.bat --list --bootstrap-server 127.0.0.1:9092
streams-plaintext-input
streams-wordcount-output
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0>
```

Starting the Wordcount java module.

```
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> set KAFKA_HEAP_OPTS=-Xmx512m
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-run-class.bat org.apache.kafka.streams.examples.wordcount.WordCountDemo ^
[2025-09-25 01:03:05,015] WARN The configuration 'admin.retries' was supplied but isn't a known config. (org.apache.kafka.clients.consumer.ConsumerConfig)
[2025-09-25 01:03:05,015] WARN The configuration 'admin.retry.backoff.ms' was supplied but isn't a known config. (org.apache.kafka.clients.consumer.ConsumerConfig)
```

Main Results:

Producer that connects to streams-plaintext-input and input typed.

```
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-console-p
roducer.bat --broker-list 127.0.0.1:9092 --topic streams-plaintext-input

>Hi Kafka this
>is a test writing
>to check if the
>program works
>running successfully
>finished fully finished
>hello world hello
>
```

Consumer that connects to streams-wordcount-output and output observed.

```
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-console-c
onsumer.bat --bootstrap-server 127.0.0.1:9092 --topic streams-wordcount-
output --from-beginning --property print.key=true --property value.deser
ializer=org.apache.kafka.common.serialization.LongDeserializer

hi      1
kafka   1
this    1
is      1
a       1
test    1
writing 1
to       1
check   1
if      1
the     1
program 1
works   1
running 1
success 1
fully   1
finishe 1
fully   1
finishe 2
hello   1
world   1
hello   2
```